Name: Grisha P
Roll No: 241901030

Experiment No. 4

# Detect Session Hijacking

## AIM

To detect session hijacking attempts on a network by analyzing network traffic for anomalies such as duplicate session cookies, simultaneous sessions from different IP addresses, and abnormal session behavior using tools such as Wireshark.

## PROCEDURE

1. Set up Wireshark on the server-side network segment to capture live traffic to and from the web application.
2. Simulate a normal session by having a legitimate user log in and browse the application; capture the baseline traffic pattern and session cookie.
3. Simulate a session hijacking attempt (reusing the captured session cookie from a different machine/IP, as in Experiment 3).
4. Apply Wireshark display filters (e.g., http.cookie, ip.addr) to isolate HTTP requests carrying the same session cookie value.
5. Compare the source IP addresses and User-Agent strings associated with requests using the same session ID.
6. Flag sessions where the same cookie value appears from two different IP addresses or two different User-Agents within a short time window as a potential hijack.
7. Cross-check server access logs for the same anomaly and record the findings.

## SAMPLE OUTPUT

```
Wireshark filter applied: http.cookie contains "PHPSESSID=8f14e45"

Frame 102  Src: 192.168.1.10 (victim)
PHPSESSID=8f14e45fceea167a5a36dedd4bea2543
Frame 215  Src: 192.168.1.25 (attacker)
PHPSESSID=8f14e45fceea167a5a36dedd4bea2543

Time difference between the two requests: 42 seconds
User-Agent (victim)   : Mozilla/5.0 (Windows NT 10.0) Chrome/118
User-Agent (attacker) : Mozilla/5.0 (X11; Linux x86_64) Firefox/119

Alert: Same session ID used from two different IP addresses and
two different browsers within 1 minute -> possible session hijacking detected.
```

## RESULT

*Session hijacking was successfully detected by analyzing network traffic in Wireshark for duplicate session identifiers originating from different IP addresses and user agents, confirming an anomalous, unauthorized reuse of a legitimate session.*